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Elongated member connector

Abstract

A connector for connecting elongated members has a connector body having two configurations. In a first configuration, the connector connects two elongated members in a fixed relative position. In a second configuration, the connector connects the two elongated members by selectively positioning one elongated member relative to a second elongated member. A threaded lock is provided that is operated to configure the connector in one of the two configurations.

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Background/Summary

FIELD OF THE INVENTION

(1) The present disclosure relates to elongated member connectors and, more particularly, to an elongated connector that can selectively position an elongated member relative to the length of another elongated member.

BACKGROUND OF THE INVENTION

(2) Connectors or couplers for connecting elongated members are known. For example, Published U.S. Patent Application No. 2016/0376811 describes a connector for securing elongated rail members together to form an assembled structure. Other connectors for connecting elongated members are described in U.S. Published US Patent Application No. 2009/0103976, and U.S. Pat. No. 11,319,975.

(3) While these and other existing connectors for connecting elongated members serve their respective objectives, a need remains for a new connector for connecting elongated members that allows for the selective positioning of one elongated member relative to a second elongated member and also allows for a fixed, non-selective position of the one elongated member relative to the second elongated member.

SUMMARY OF THE INVENTION

(4) Embodiments of the invention provide a connector for connecting elongated members that permit the selective positioning of one elongated member relative to a second elongated member.

(5) Embodiments of the invention further provide a connector for connecting elongated members that has two operating configurations. In the first configuration, two elongated members are connected in a fixed relative relation, and in the second configuration, one elongated member may be selectively positioned related to the other elongated member. In other words, in the first configuration, the connector is configured like an elbow connector, and in the second configuration, the connector can slide up and down one of the elongated members.

(6) Embodiments of the invention further provide a connector for connecting elongated members of the frame elements of game net frames that are used to hold a game net.

(7) Numerous additional objects, features and advantages of the present invention will be readily apparent to those of ordinary skill in the art upon a reading of the following detailed description of presently preferred, but nonetheless illustrative, embodiments of the present invention when taken in conjunction with the accompanying drawings. The invention is capable of other embodiments and of being practiced and carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein are for the purpose of descriptions and should not be regarded as limiting.

(8) As such, those skilled in the art will appreciate that the conception, upon which this disclosure is based, may readily be utilized as a basis for the designing of other structures, methods, and systems for carrying out the several purposes of the present invention. It is important, therefore, that the claims be regarded as including such equivalent constructions insofar as they do not depart from the spirit and scope of the present invention.

(9) For a better understanding of the invention, its operating advantages and the specific objects attained by its uses, reference should be had to the accompanying drawings and descriptive matter in which there are illustrated embodiments of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following drawings illustrate by way of example and are included to provide further understanding of the invention for the purpose of illustrative discussion of the embodiments of the invention. No attempt is made to show structural details of the embodiments in more detail than is necessary for a fundamental understanding of the invention, the description taken with the

drawings making apparent to those skilled in the art how the several forms of the invention may be embodied in practice. Identical reference numerals do not necessarily indicate an identical structure. Rather, the same reference numeral may be used to indicate a similar feature or a feature with similar functionality. In the drawings:

- (2) FIG. 1 is a first perspective view of an elongated member connector that is constructed in accordance with the principles of an embodiment of the invention;
- (3) FIG. 2 is a second perspective view of an elongated member connector that is constructed in accordance with the principles of an embodiment of the invention;
- (4) FIG. 3 is a top view of an elongated member connector that is constructed in accordance with the principles of an embodiment of the invention;
- (5) FIG. 4 is a cross-sectional view taken along line 4-4 in FIG. 1;
- (6) FIG. 5 is an exploded perspective view of an elongated member connector that is constructed in accordance with the principles of an embodiment of the invention;
- (7) FIG. 6 is a side view of an elongated member connector that is constructed in accordance with the principles of an embodiment of the invention, shown connecting two elongated members;
- (8) FIG. 7 is a side cross-sectional view of an elongated member connector that is constructed in accordance with the principles of an embodiment of the invention, shown connecting two elongated members and with the connector in a first configuration;
- (9) FIG. 8 is a side cross-sectional view of an elongated member connector that is constructed in accordance with the principles of an embodiment of the invention, shown connecting two elongated members and with the connector in a second configuration;
- (10) FIG. 9 is an illustration of a game net frame having elongated frame members connected by elongated member connectors that are constructed in accordance with the principles of an embodiment of the invention, showing the elongated member connectors in a first configuration; and
- (11) FIG. 10 is an illustration of a game net frame having elongated frame members connected by elongated member connectors that are constructed in accordance with the principles of an embodiment of the invention, showing the elongated member connectors in a second configuration.

DETAILED DESCRIPTION

(12) The following detailed description of embodiments of the invention references the accompanying drawings. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention.

(13) Turning now to FIGS. 1-5 of the drawings, an elongated member connector **10** that is constructed in accordance with an embodiment of the invention is illustrated. Connector **10** has a body **12** that is configured to connect two elongated members (not shown). As representatively shown, body **12** is configured to connect two elongated members at a generally 90-degree angle to one another. The connector body **12** can be formed as an integral one piece from a plastic material. As further discussed herein, the elongated members could be elements of a frame that is used to support a game net. It is important to note the elongated members do not need to be hollow tube-like members. Rather, the elongated members could be a solid rod, a rail, a pipe, post, or a tubular member, for example.

(14) As representatively shown, body **12** has first and second passages **14** and **16** that are arranged transversely to one another and can be at a 90-degree angle relative to one another. As further representatively shown, the first and second passages **14** and **16** are cylindrical and configured to receive a cylindrically shaped elongated member. However, it should be noted that the first and second passages **14** and **16** could have different cross-sectional geometries that correspond to different shapes of elongated members. As a nonlimiting example, the first and second passages **14** and **16** could be square shaped in cross-section to accommodate similarly square shaped elongated members.

(15) The first passage **14** is open at opposite ends **18** and **20** and is configured to slidably receive a

first elongate member such that the body **12** is slidable along a length of the elongated member. The second passage **16** is open at end **22** and is configured to receive a second elongated member. The second passage **20** can be a blind bore such that the insertion depth of the second elongated member may be limited by end wall **24**.

(16) The connector **10** further includes a lock **26**. As representatively shown, lock **26** has a hand knob **28** having a stop surface **30** and a shaft **32**. Shaft **32** extends outwardly from the stop surface **30** and that terminates at free end **34**. Shaft **32** has a threaded section **36** and a non-threaded stub section **38** that is located at the free end. The shaft **32** is received by a threaded bore **40** of the body **12** with the threaded section **36** threadedly engaged with the threads of the threaded bore. The threaded bore **40** can at least be partially formed through boss **42** of body and the threaded bore extends through the body and is open to the first passage **14**.

(17) Rotating the knob **28** in a first direction advances the shaft **32** inwardly causing the non-threaded stub section **38** to extend out of the threaded bore and transversely into the first passage **14**. Rotating the knob **28** in a second direction opposite of the first rotational direction causes the shaft to move outwardly and withdraws the non-threaded stub section **38** from the first passage.

(18) Continued rotation in the first direction causes the stop surface **30** to abut against the body **12**, and as representatively shown, abut against surface **44** of the boss **42** to prevent any portion of the threaded section **36** from extending out of the threaded bore and into the first passage **14**. In an aspect, the distance between stop surface **30** and the position where the threaded section **36** transitions to the non-threaded stub section **38** is equal to or less than the length of the threaded bore **40**. As explained in more detail below, this full stop position prevents the threads of the threaded section **36** from being damaged by an elongated member disposed in the first passage.

(19) The body **12** may further have a flange **46** extending therefrom and a hole **48** through the flange. The hole **48** can be used to attach a tie rope or guy wire in certain applications, such as, for example when the connector is used in the assembly of a net frame and the tie rope is used to help secure the net frame.

(20) In FIG. **6** the connector **10** is shown with a first elongated member **50** received by the first passage **14** and a second elongated member **52** received by the second passage **16**, thereby connecting the first and second elongated members.

(21) In FIG. **7** the connector **10** is shown in cross-section and in a first configuration with the first and second elongated members **50** and **52** received by the first and second passages **14** and **16**, respectively. Further shown is the lock **26** rotated into the full stop position with the stop surface **30** of the lock abutting against surface **44** of the body **12**. In this position, the non-threaded stub section **38** of shaft **32** extends into the first passage **14** and provides an abutment against end **54** of the first elongated member **50**. In such a configuration, the first elongated member **50** is not permitted to extend completely through the first passage **14**, thereby positioning the connector **10** at the end of the first elongated member.

(22) As further shown, the full stop position of the lock **26** prevents the threaded section **36** of shaft **32** from extending into the first passage **14** such that the end **54** of the first elongated member cannot contact the threaded section which prevents the end from damaging the threads.

(23) In FIG. **8** the connector **10** is shown in a second configuration. Here, the lock **26** is positioned with the non-threaded stub section **38** of shaft **32** withdrawn from the first passage **14**, which allows the body **12** to slide along the length of the first elongated member **50**. Sliding the body **12** along the first elongated member **50** also moves the position of the second elongated member **52** relative to first elongated member, thereby adjusting the coupling position between the two members. The position of body **12** along the first elongated member **50** can be selectively retained by rotating the shaft **32** of lock **26** in the first direction to cause the shaft end **34** to abut against the exterior surface of the elongated member and clamp the body to the member.

(24) In FIG. **9** two connectors **10** are shown in use connecting frame members **56**, **58**, and **60** of a game net frame **72** upon which a game net **74** is attached. As shown one connector **10** connects

vertical member **56** and horizontal member **60**, and the other connector **10** connects vertical member **58** and the horizontal member. Here, connectors **10** are in the first configuration as shown in FIG. 7 and explained above, wherein the connectors are positioned at the ends of members **56** and **58**.

(25) In FIG. **10** the game net frame **72** shown in FIG. **9** is further illustrated with the connectors **10** in the second configuration, which is shown in FIG. **8** and explained above. In this configuration, connectors **10** are slid in a direction downward along a length of the vertical members **56** and **58** which also moves the horizontal member **60** downward in relation to the vertical members. With the top edge of the net **74** attached to the horizontal member **60**, the height of the net can be adjusted by adjusting the position of connectors **10** along the vertical members **56** and **58**.

(26) It will be appreciated by persons skilled in the art that the present embodiment is not limited to what has been particularly shown and described hereinabove. A variety of modifications and variations are possible in view of the above teachings without departing from the scope of the disclosure.

Claims

1. A connector for joining elongated members, the connector comprising: a connector body; the connector body having a first passage with open opposite ends for receipt of a first elongated member, the first passage configured such that the connector body is slidable along a length of the first elongated member; the connector body having a second passage with an open end for receipt of an end of a second elongated member; a lock having a stop surface and a shaft terminating at a free end, the shaft having a threaded section and a non-threaded stub section at the free end, the threaded section of the shaft being threadedly received by a threaded bore of the connector body such that the non-threaded stub section is extended into the first passage by rotating the shaft in a first direction and retraced from the first passage by rotating the shaft in a direction opposite of the first direction; and wherein the threaded section of the shaft is prevented from extending into the first passage by the stop surface abutting against the connector body.
 2. The connector of claim 1, wherein the first and second passages are transverse to one another.
 3. The connector of claim 1, wherein the first and second passages are arranged at 90-degrees to one another.
 4. The connector of claim 1, wherein the lock further comprises a hand knob.
 5. The connector of claim 1, wherein the connector body further comprises a flange and a hole through the flange.
 6. The connector of claim 1, wherein the first and second passages are cylindrical.
 7. The connector of claim 1, wherein the connector body is integrally formed as one piece from a plastic material.
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